

MECHTRON 4X03: Scientific Computation

Course Outline and Syllabus

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Introduction

Scientists conduct experiments to ask interesting and challenging questions about the physical world. Similarly, engineers design solutions to real-world problems involving a variety of mechanical, electrical, and chemical systems and processes. In most cases, the best models of natural phenomena are *continuous*, and we must use real numbers to describe and predict their behaviour. For example, Newton’s second law in one dimension is

$$F(x, \dot{x}, \ddot{x}, t) = ma = m\ddot{x}, \quad (1)$$

which describes a second-order ordinary differential equation. Depending on the properties of the force function F , Equation 1 may or may not have an analytic solution (i.e., one that can be worked out with pen and paper). In this course, we will learn techniques for solving problems from the physical sciences¹ with computers (e.g., problems involving Equation 1 that cannot be solved in closed form). As we study numerical methods, we will find that straightforward translation of mathematical formulae and definitions into software is often not possible. This difficulty arises from numerous sources, including but not limited to:

1. the nonlinearity of real-world problems;
2. resource and hardware constraints (e.g., limited memory and computation time);
3. complexity of algorithms that carry out advanced mathematical operations; and
4. challenges arising from discrete (i.e., digital) representations of continuous (i.e., real) numbers.

The tools we will use to explore and overcome these issues can be understood with elementary linear algebra and calculus. In fact, the main “trick” we will use throughout the course is simply the Taylor series approximation of nonlinear functions of interest. Scientific computation is a unique topic at the intersection of theory and practice, and assessments will involve both programming and mathematical analysis.

Teaching Team

Instructor

Matthew Giamou, giamoum@mcmaster.ca

¹Although this course’s focus is on problems relevant to mechatronics and software engineering, the methods we will develop are applicable to the solution of *any* problem involving real numbers, including those arising in pure math, operations research, biology, and the social sciences.

Teaching Assistants

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Schedule

Lectures

Monday, 1:30pm, JHE 376
Wednesday, 1:30pm, JHE 376
Thursday, 1:30pm, JHE 376

Tutorials

Since there is a graded quiz component to most tutorials and limited space in each room, you are expected to only attend your assigned tutorial section.

Tuesday, 10:30am, ETB 238 (Section 1)
Monday, 9:30am, JHE 326H (Section 2)
Wednesday, 10:30am, JHE A101 (Section 3)
Thursday, 4:30pm, ETB 235 (Section 4)

Office Hours

Office hours with Professor Giamou will be announced in class.

Learning Objectives

The successful student will be able to:

1. summarize the advantages and disadvantages of floating-point representations of real numbers;
2. detect and fix issues in floating-point computations (e.g., overflow, cancellations, and roundoff errors);
3. derive error bounds on computational results;
4. select and implement an appropriate solution method for a variety of linear systems, interpolation problems, unconstrained continuous optimization problems, and ordinary differential equations;
5. describe and implement various numerical integration schemes;
6. apply linear algebraic concepts to solving and characterizing linear least squares problems;
7. perform simple complexity analysis on numerical algorithms;
8. explain the importance of eigenvalues and eigenvectors in numerical methods;
9. analyze the convergence and stability of numerical routines; and
10. write Julia programs implementing numerical methods.

Materials

There is no required textbook as all course material will be covered in class and subsequently posted on Avenue to Learn. However, Kyle Novak's "Numerical Methods for Scientific Computing" is [available for free online](#) and is a useful supplementary reference, especially for example Julia code. Additionally, chapters from the sixth edition of Cheney and Kinkaid's "Numerical Mathematics and Computing" are listed alongside each section in the course outline and may be a useful supplement to lecture material. Finally, students who need a refresher on linear algebra will find "Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares" by Stephen Boyd and Lieven Vandenberghe helpful (it also serves as a useful companion to lecture material in weeks three through six). The instructor may occasionally recommend additional supplementary readings: these will be announced in class and made available on Avenue to Learn. Practice problems will be drawn from a variety of online and print sources and cited appropriately. Students are encouraged to explore these sources for further practice material or to satisfy their curiosity.

Syllabus

The schedule of course topics in this section is a rough guide and subject to change as the semester progresses. In general, the week's lectures will cover new material in depth and also involve solving examples of mathematical/analytical problems. Tutorials will focus on solving programming problems with the Julia language and should lag behind lecture material by about one week. Lectures and tutorials will **not** be recorded, but all notes will be posted on Avenue. Once again, the sections from Cheney and Kinkaid's textbook (abbreviated CK) listed each week are optional - we will develop all mandatory material in lecture, tutorials, and assignments.

Week 1: Sept. 2–6

First lecture is Wednesday, September 4th. Introduction to scientific computation. Example applications. Course outline, policies, procedures, and goals. Scheduling office hours. Introduction to floating- and fixed-point arithmetic. Rounding. **No tutorial.**

CK Readings: §1.1

Week 2: Sept. 9–13

Machine precision. IEEE 754 standard. Floating point arithmetic. Cancellations. Examples. **Tutorials begin. Assignment 1 released.**

CK Readings: §2.1, §2.2

Week 3: Sept. 16–20

Calculus preliminaries (Taylor series). Truncation, roundoff, and computational error. Linear algebra review. Linear systems and their direct solution (i.e., Gaussian elimination).

CK Readings: §1.2, §4.3, §7.1

Week 4: Sept. 23–27

Partial pivoting. LU decomposition. Error analysis in the solution of linear systems. Condition numbers.

CK Readings: §7.2, §8.1

Week 5: Sept. 30 – Oct. 4

Structured linear systems. Thomas algorithm for tridiagonal systems. QR and Cholesky factorizations. Iterative solutions (i.e., the Jacobi method).

CK Readings: §7.3, §8.1, §8.2

Week 6: Oct. 7–11

Introduction to linear least squares problems. Linear algebraic analysis. Overdetermined systems. Underdetermined systems and the Moore-Penrose pseudoinverse. Normal equations. **Assignment 1 due. Assignment 2 released.**

CK Readings: §12.1

Week 7: Oct. 14–18

Mid-term recess - no lectures or tutorials.

Week 8: Oct. 21–25

Computing eigenvalues and eigenvectors with the power method and the Rayleigh quotient. Bauer-Fike bounds. **Midterm.**

CK Readings: §8.3, §8.4

Week 9: Oct. 28 – Nov. 1

Root finding. Newton's method for nonlinear equations. Introduction to unconstrained continuous optimization. First- and second-order optimality. Rate of convergence.

CK Readings: §3.1, §3.2,

Week 10: Nov. 4–8

Introduction to interpolation. Basis functions. Polynomial (Lagrange) interpolation. Newton's form of polynomial interpolation. Errors in polynomial interpolation. Chebyshev nodes. Splines if time permits.

CK Readings: §4.1, §4.2

Week 11: Nov. 11–15

Introduction to numerical integration. Trapezoidal, midpoint, and Simpson's rule. Composite rules. Adaptive Simpson's rule. Error analysis. **Assignment 2 due. Assignment 3 released.**

CK Readings: §5.1, §5.2, §6.1

Week 12: Nov. 18–22

Introduction to numerical solution of ordinary differential equations (ODEs). Euler's method. Stability analysis. Error analysis. Convergence for ODEs. Numerical stiffness. Runge-Kutta methods.

CK Readings: §10.1, 11.1

Week 13: Nov. 25–29

Numerical differentiation. Boundary value problems. Guest lecture. **Assignment 3 due.**

CK Readings: §10.2, §10.3

Week 14: Dec. 2–6

Discussion of applications and review.

	Weight (%)	Released	Due
Tutorial Quizzes	15	Weekly	In tutorial
Assignment 1	10	Friday Sept. 13	Friday Oct. 11
Midterm	20	Wednesday Oct. 23	Wednesday Oct. 23
Assignment 2	10	Friday Oct. 11	Friday Nov. 8
Assignment 3	10	Friday Nov. 8	Friday Nov. 29
Final Exam	35	TBA	TBA

Table 1: Breakdown of marks.

Grading

Marks are summarized in Table 1 and will come from three sources:

1. take-home assignments;
2. midterm and final exams focused on mathematical analysis conducted without a computer (dates to be announced); and
3. weekly tutorial quizzes.

Tutorial quizzes will take place over the final 15-30 minutes of each weekly tutorial session. Any tutorials which you are unable to attend will have their grade weight automatically rolled into the final exam (there is no need to use an MSAF or inform me if you miss a quiz). The three take-home assignments are **individual assessments**. The midterm is closed-book with the exception of one double-sided 8.5x11" sheet of notes which may be digitally printed or hand-written. Likewise, the final is closed-book with the exception of **two** double-sided 8.5x11" sheets of notes which may be digitally printed or hand-written. You are permitted to use a standard McMaster calculator to both the midterm and final exam.

Remarking

1. Requests for remarking of an assignment or a test must be made within one week after the marked assignment/test is returned.
2. Requests that are later than a week will not be accommodated.

Course Policies

Please read and understand the policies outlined in this section. If you are confused or have any questions about a policy, please ask the instructor.

Absences and Late Work

1. Late work will not be graded without an MSAF.
2. The MSAF accommodation for a missed assignment is a **six calendar day** extension from the original assignment deadline.
3. The MSAF accommodation for a missed midterm is to roll the weight of the assessment into the weight of the final examination.
4. Missing a quiz **automatically** rolls the quiz's weight into the weight of the final examination; do not attempt to use an MSAF for this purpose.

Please review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: Grade of F assigned for academic dishonesty), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

Use of Generative AI

Students may use generative artificial intelligence (AI) in this course in accordance with the guidelines outlined for each assessment, and so long as the use of generative AI is referenced and cited following citation instructions given in this syllabus. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the students responsibility to be clear on the limitations for use for each assessment and to be clear on the expectations for citation and reference and to do so appropriately.

Disclaimer

It is the opinion of your instructor that you should not use generative AI tools on any of the assignments. Scientific computation is a detail-oriented subject combining abstract mathematics, complex algorithms, and esoteric knowledge of software and hardware. Students will be best served if they develop the skills and knowledge to independently implement and debug numerical methods in a variety of contexts. As novices, learning this craft will involve the solution of simpler problems that are illustrative of key principles, but not as complicated as real-world examples, and therefore may be within a large language model's capabilities. Generative AI is a powerful tool, but it is not prudent for serious scholars to “short circuit” their learning for the sake of convenience. Your take-home assignments will have further instructions on permissible use of generative AI, and failure to comply with these guidelines will result in a mark of zero, and in some instances may constitute a breach of academic integrity.

Citation Instructions

When citing traditional academic sources, use a consistent style that is as close to [IEEE-style citations](#) as is reasonably convenient. For assignments which permit you to make use of text-based generative AI, please adhere to the following procedure:

1. Save a text transcript or screenshot of *all* prompts and outputs generated while working with generative AI tools on the project.
2. Include this record of your interactions with your submission.
3. Throughout the rest of your submission, clearly reference any elements, including code, equations, and prose, which were drawn from or inspired by your interactions with generative AI tools. This can be conveniently achieved with a reference like the following at the end of your submission:

ChatGPT-3.5 and M. Giamou, *Help me understand rounding error*. Interaction with generative AI tool (August 16, 2024).

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities policy](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances

Students requiring academic accommodation based on religious, indigenous or spiritual observances (RISO) should follow the procedures set out in the [RISO policy](#). Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Online Elements

This course uses online elements (e.g. email, and Avenue to Learn). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster email accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses online elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn, and/or McMaster email.

Changes

The instructor reserves the right to modify elements of this course and will notify students accordingly.

Acknowledgements

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